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A mystery tale: Nickel is fickle when snails fail. Investigating the variability in Ni toxicity to the great pond snail.

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DISCLAIMER

The authors declare no conflicts of interest.

DATA AVAILABILITY

Details of the statistical analyses (Tables SI.1 and SI.2), BLM calculations (Table SI.3) and all the data illustrated in the present paper (Tables SI.4 to SI.8) are provided in a Supplemental Information file. Figure SI.1 compares the J_{int} values reported in Niyogi et al. (2014) and Crémazy et al (2019). Figure SI.2 shows the effect of test duration on snail SGR in Exp#3 and Niyogi et al. (2014). Figure SI.3 shows the effect of snail age at toxicity test start on $EC_{20\text{SGR}}$ BLM residuals (calculated from Table 2) of the four published Ni studies.

Abstract

Dissolved nickel (Ni) concentrations inhibiting the growth of juvenile great pond snails (*Lymnaea stagnalis*) have been documented to vary from about 1 to 200 $\mu\text{g L}^{-1}$ Ni. This variability makes *L. stagnalis* either a moderately sensitive or the most sensitive freshwater species tested to date to chronic Ni exposure. Given the role of sensitive species in environmental risk assessment frameworks, it is particularly important to understand this variability, *i.e.* to characterize the factors that modulate Ni toxicity and that may confound toxicity test outcomes when uncontrolled. In the present study, we tested if this variability is due to analytical (growth calculation: biomass *versus* growth rate), environmental (water quality), lab-specific practices and/or source population differences among earlier studies. Specifically, we re-analyzed previously published Ni toxicity data and performed additional measurements of Ni aqueous speciation, short-term Ni uptake and chronic Ni toxicity with test waters and snail cultures used in previous studies. Corrections for nickel bioavailability and growth calculations explained a large degree of variability in the published literature. However, a residual 16-fold difference remained puzzling between two studies: Niyogi et al. (2014) (low ECxs) and Crémazy et al. (2018) (high ECxs). Indeed, differences in metal bioavailability due to water chemistry, lab-specific practices, and in snail population sensitivity could not explain the large variation in Ni toxicity in these two very similar studies. Other potentially important toxicity-modifying factors were not directly evaluated in the present work: test duration, diet, snail holding conditions and snail age at onset of testing. The present analysis highlights the need for further studies to elucidate i) the mechanisms of growth inhibition in Ni-exposed *L. stagnalis* and ii) the important abiotic and biotic

factors affecting this biological response. Until these processes are understood, substantial uncertainties will remain about inclusion of this species in Ni environmental risk assessment.

Keywords: chronic toxicity, *Lymnaea stagnalis*, freshwater, bioavailability, growth inhibition

INTRODUCTION

The great pond snail *Lymnaea stagnalis* is an abundant and widespread pulmonate snail present in temperate fresh and brackish waters in North America, Europe, Asia and North Africa (Amorim et al. 2019). As part of the diet of many species (fish, crustaceans, amphibians, mammals and birds), it is considered a key organism for ecosystem health and function (Bronmark 1994; Nyström and Pérez 1998). Since the early 2000s, *L. stagnalis* has been increasingly used as a model organism in ecotoxicology studies, mainly because the snails are easy to culture in the lab, ecologically important, and highly sensitive to pollutants (Amorim et al. 2019). Regarding this latter aspect, the extreme sensitivity of *Lymnaea stagnalis* to long-term exposures to many trace metals also makes it a key organism for the environmental regulation of these inorganic contaminants. Indeed, over the past decade, the great pond snail was shown to be either the most-sensitive, or the second-most sensitive organism to Co, Cu, Ni and Pb in chronic exposures (Grosell et al. 2006; De Schamphelaere et al. 2008; Schlekot et al. 2010; Brix et al. 2011; Niyogi et al. 2014). In these studies, effects on juvenile growth were observed at low $\mu\text{g L}^{-1}$ aqueous metal concentrations, which are close to protective values for aquatic life. For example, Brix et al. (2011) measured a 30-d EC20 of $1.8 \mu\text{g L}^{-1}$ Cu for

juvenile snail growth, indicating that *L. stagnalis* is under-protected by the USEPA hardness-based Cu water quality criteria ($7.8 \mu\text{g L}^{-1}$ under the test conditions). While the exact mechanisms of metal toxicity remain elusive, this high sensitivity seems to be linked to disruption of the homeostasis of Ca^{2+} , a substrate that sustains the extremely fast shell formation required for high growth rates of juvenile *L. stagnalis* (~20 - 25% biomass gain per day) (Grosell and Brix 2009; Brix et al. 2012). Nickel may also disturb Na^+ , Mg^{2+} and Fe^{2+} homeostasis (Leonard and Wood 2013; Niyogi et al. 2014; Brix et al. 2017). These data suggest that this sensitive organism may have an important role in revisions of current water quality guidelines. Yet, conflicting data on *L. stagnalis* sensitivity have recently emerged for nickel (Ni) that create uncertainty for how to incorporate data from this species in nickel risk assessments.

Lymnaea stagnalis has been reported to be among the most chronically sensitive species tested to date for Ni. Using newly hatched (< 24-h old) snails, Schlekert et al. (2010) measured 30-d EC20s for growth ranging from 1.6 to $21 \mu\text{g L}^{-1}$ in five natural waters representing the typical range of water composition in temperate regions. This variability could be mainly explained by differences in water chemistry – *i.e.* by changes in Ni bioavailability, using a Biotic Ligand Model (BLM). When these toxicity data were normalized to a common water chemistry, *L. stagnalis* and *Ceriodaphnia dubia* were found to be the two most sensitive species (with comparable sensitivities) in the Ni ecotoxicity database (European Chemicals Bureau 2008). By comparison, the species in the middle of the list of sensitivity, the green alga *Pediastrum duplex* is approximately 16 times less sensitive than *L. stagnalis* and 14 times less sensitive than *C. dubia*. Other

growth inhibition tests supported these findings: Niyogi et al. (2014) reported a 21-d EC20 < 1.3 $\mu\text{g L}^{-1}$ (lowest tested concentration) using one-week old snails and Nys et al. (2016) reported 28-d EC20s of 10 and 32 $\mu\text{g L}^{-1}$ using 2- or 3-d old snails in waters of higher pH and hardness. These lab results were further corroborated in the field by Peters et al. (2014), with effects on snail abundance occurring at lower levels (3.9 $\mu\text{g L}^{-1}$ bioavailable Ni) than on other sensitive benthic macroinvertebrates (Ephemeroptera, Plecoptera or Tricoptera). Additionally, Hommen et al. (2016) showed that *Lymnaea* was the most sensitive species out of >150 species exposed to Ni in a 12 week microcosm study. Meanwhile, a more recent study found *L. stagnalis* to be much more tolerant to Ni, with a 14-d EC20 of 150 $\mu\text{g L}^{-1}$ using 2 – 3 week old snails (Crémazy et al. 2018). The various Ni toxicity lab studies were carried out using different test protocols and conditions (Table 1), in part due to the absence of a standardized method for testing growth inhibition in *L. stagnalis*. Yet, no obvious reason explains the high variability in Ni toxicity, which ranges from 1 to 200 $\mu\text{g L}^{-1}$ approximately. Notably, test conditions (including water chemistry) used in Crémazy et al. (2018) were very similar to those used in Niyogi et al. (2014). Interestingly, Crémazy et al. (2018) measured EC20s for five other metals that were similar to other published data: silver (Brancho 2017), cadmium (Pais 2012), copper (Brix et al. 2011; Brancho 2017), lead (Brix et al. 2012) and zinc (De Schamphelaere and Janssen 2010). Thus, this sensitivity variability seems to be a Ni-specific issue.

In the present study, we investigated the large discrepancies reported in the literature for chronic Ni toxicity to *L. stagnalis*, with a focus on the Crémazy et al. (2018)

and the Niyogi et al. (2014) studies. These two studies were selected mainly because their test conditions were the closest (Table 1), yet their reported Ni EC20s were the most different (Table 2). Four potential factors were tested to explain the variability: i) differences in methods used to calculate the toxicity endpoint, ii) differences in Ni bioavailability in the test waters (water chemistry effects), iii) differences in sensitivity of the snail cultures (population effect) and differences in lab-specific practices (lab effect). To test these factors, we re-analyzed the literature data (Tables 1 and 2) and performed additional physiological and toxicological measurements using the two snail populations and the two test waters used in Crémazy et al. (2018) and in Niyogi et al. (2014) (Table 3). These additional experiments were performed at the lab sites where these original studies were carried out, *i.e.* at the University of British Columbia (UBC, Vancouver, Canada) and at the University of Miami (UM, Miami, FL, USA), with local and shipped (from the other lab) snails and test waters. We measured Ni²⁺ aqueous concentrations (with an ion-exchange technique), short-term Ni uptake fluxes (3-h flux measurements with radiolabeled Ni) and chronic Ni toxicity (7-d and 14-d growth tests) in juvenile snails.

The present study addresses an issue of particular relevance to the regulatory community. Indeed, accurate risk management requires a clear understanding of the factors affecting toxicity variability and test reproducibility with key sensitive species. While the present study could not explain the full extent of the variability in snail sensitivity to Ni, it provides some answers, raises important considerations for snail

toxicity testing, and makes recommendations for subsequent studies and future development of standardized methods.

MATERIALS AND METHODS

Literature data analyses

To our knowledge, only four studies have reported effects of Ni on growth of juvenile *L. stagnalis* (Table 1). These studies did not estimate effect concentrations on growth (ECx) with the same method: Schlekot et al. (2010) and Niyogi et al. (2014) used snail biomass (ECx_{biomass}) while Nys et al. (2016) and Crémazy et al. (2018) used snail specific growth rate (ECx_{SGR}) (Table 2). For each study, we calculated EC20s and EC50s based on biomass (final snail weight or shell length) and specific growth rate (SGR), to allow for inter-study comparison. The measurement of both response variables is traditionally proposed in standardized methods, as there is still no consensus on which variable should be used for risk assessment (OECD 2011). SGR (in d⁻¹) is calculated using the following equation:

$$\text{SGR} = \frac{\ln(w_f) - \ln(w_i)}{t} \quad (\text{eq.1})$$

Where w_f and w_i are respectively the initial and final snail wet weight (soft tissue + shell) (Som Niyogi, University of Saskatchewan, personal communication, Chris Schlekot, NiPERA, personal communication, Charlotte Nys, Arche Consulting, personal communication) and t is the test duration. The ECx_{SGR} in Niyogi et al. (2014) and in Schlekot et al. (2010) and the ECx_{biomass} in Nys et al. (2016) and Crémazy et al. (2018)

were then estimated with the methods used for the original ECx estimates. That is, EC_{xSGR} in Niyogi et al. (2014) were estimated by graphical interpolation, while EC_{xSGR} in Schlekot et al. (2010) and EC_{xbiomass} in Nys et al. (2016) and Crémazy et al. (2018) were estimated from the Toxicity Relationship Analyses Program (TRAP, Version 1.10, USEPA, NHERL, Duluth, MN, USA) using logistic equations, after log transformation of dissolved Ni concentrations. The logistic equation had three parameters in Schlekot et al. (2010) and Crémazy et al. (2018) and two parameters in Nys et al. (2016).

To evaluate test water chemistry effects (*i.e.* metal bioavailability changes) in these studies, we predicted the ECx in each test water using BLMs parameterized by Schlekot et al. (2010) and Nys et al. (2016). These studies used an existing chronic Ni BLM for *Ceriodaphnia dubia*, with adjusted sensitivity parameters for *L. stagnalis*, to obtain ECxs based on bioavailable Ni²⁺ concentration:

$$\text{Predicted ECx}^{\text{Ni}^{2+}} = 10^{-(Q_{x,\text{Lymnaea}} + S_{\text{pH}} \times \text{pH})} \times \{1 + K_{\text{CaBL}} \times [\text{Ca}^{2+}] + K_{\text{MgBL}} \times [\text{Mg}^{2+}]\} \quad (\text{eq. 2})$$

With $K_{\text{CaBL}} (= 10^{3.53})$ and $K_{\text{MgBL}} (= 10^{3.57})$ the biotic ligand binding stability constants for Ca^{2+} and Mg^{2+} respectively, $S_{\text{pH}} (= 0.8587)$ the slope of the pH effect, and $Q_{x,\text{Lymnaea}}$ the intrinsic sensitivity parameter for x% effect on *L. stagnalis* growth. Schlekot et al. (2010) derived a $Q_{50, \text{Lymnaea}, \text{biomass}}$ of 1.099 for prediction of $\text{EC50}^{\text{Ni}^{2+}}_{\text{biomass}}$. Nys et al. (2016) derived a $Q_{20, \text{Lymnaea}, \text{SGR}}$ of 0.29 for prediction of $\text{EC20}^{\text{Ni}^{2+}}_{\text{SGR}}$. The corresponding $\text{EC50}_{\text{biomass}}$ and EC20_{SGR} based on total Ni concentrations were then obtained with the Windermere Humic Aqueous Model (WHAM, version VII, (2011)), following the modeling recommendations from Van Laer et al. (2006), as in Schlekot et al. (2010).

Experimental water compositions

We conducted additional measurements of Ni speciation, uptake flux rate and chronic toxicity in fresh test waters used in Niyogi et al. (2014) at the University of Miami (UM) and in Crémazy et al. (2018) at the University of British Columbia (UBC) (Table 3). In the remainder of this manuscript, these two waters will be referred to as “UM water” and “UBC water” respectively. In some cases, test water was shipped from UBC to UM and vice-versa.

The UM water corresponds to dechlorinated Miami city water. Its composition is given in Niyogi et al. (2014), but the concentrations of major cations, dissolved organic carbon (DOC) and dissolved inorganic carbon (DIC) were also re-measured during the present study (Table 4). These chemistry parameters were not noticeably different between the two studies. The UBC water corresponds to dechlorinated Vancouver city water, to which salts were added ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$, NaHCO_3 , KHCO_3 , NaCl and $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$) to mimic the ionic composition of Miami water (although Ca and SO_4 levels in UBC water were respectively 2- and 6-fold greater). Its composition was measured in Crémazy et al. (2018) and is also given in Table 3. There were two reasons for this salt addition to Vancouver water. First, this water is very soft and snails perform better in moderately hard to hard waters. Second, a series of metal chronic toxicity tests had already been performed at UM (Cu: Brix et al. 2011; Pb: Brix et al., 2012; Ni: Niyogi et al.; 2014) and they could provide a useful comparison to the UBC tests carried out in Crémazy et al. (2018, 2019) if conducted in relatively similar water chemistry. The main

differences between the two waters were their Ca (2-fold higher in UBC water), DOC (3-fold higher in UM water) and SO₄ concentrations (6-fold higher in UBC water).

Experimental snail culture conditions

The additional Ni uptake flux rate and chronic toxicity tests were conducted with two snail cultures: i) the culture used by Crémazy et al. (2018, 2019) at UBC and ii) the culture used by Niyogi et al. (2014) at UM (Table 3). In some cases, egg masses were shipped from UBC to UM and vice-versa.

The UBC snails have been continuously cultured since 2014, from a colony held at McMaster University since 2006. The McMaster culture is the result of the mixing of five cultures from the following sources: N. Syed (University of Calgary, Calgary, Alberta, Canada). M. Grosell and S. Ebanks (University of Miami, Florida, USA), Z.-P. Feng (University of Toronto, Toronto, Ontario, Canada), G. Spencer (Brock University, St. Catharines, Ontario, Canada) and D. Spafford (University of Waterloo, Waterloo, Ontario, Canada). Thus, the UM culture and the McMaster/UBC culture have some degree of relatedness. At UBC, snails were kept in UBC water, at 25 ± 1 °C, with a 16h light: 8h dark photoperiod. They were fed with excess peels of washed sweet potato (*Ipomoea batatas*) for newly-hatched snails to approximately 1 - 2 week old snails, then with washed romaine lettuce (*Lactuca sativa*), three times a week. Holding tank water was changed twice a week. At UM, the snails have been continuously cultured since 2002. The original source of this culture was Dr. N. Syed at the University of Calgary – also one of the sources of the McMaster/UBC colony; thus the two cultures have been separated for about a decade. The UM culture conditions were the same as described

above for UBC except adult snail cultures were maintained under flow-through conditions using UM water.

Egg masses for experiments were isolated and maintained under static renewal conditions through hatch and until use in toxicity tests. Shipped egg masses were acclimated to the local culture conditions until tests were initiated. They were provided sweet potato until ~10 days post hatch, then a mix of sweet potato and lettuce. Snails were primarily feeding on lettuce when Ni uptake and toxicity tests were initiated (2 – 4 weeks post-hatch). These tests were performed under the same temperature and light regime as described above.

Short-term Ni uptake tests

Short-term Ni uptake rates were measured in UBC snails exposed in parallel in UM water or in UBC water (Exp#1 in Table 3), as a physiological indicator of Ni bioavailability in both waters. These tests were performed at UBC, as previously detailed in Crémazy et al. (2019). Exposure solutions were prepared 24 h prior to the test. They consisted of test waters spiked with $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ (ACS grade, Fisher Scientific) at 40 and 300 $\mu\text{g L}^{-1}$ Ni, and with 0.02 $\mu\text{Ci mL}^{-1}$ of radioactive ^{63}Ni (Stuart Hunt & Associates Ltd, Edmonton, AB, Canada). Snails approximately 4 weeks old (100 - 150 mg wet weight) were selected for these tests. They had been switched from a sweet potato to a lettuce diet about 1 week before the test and were starved for 24 hours prior to the exposure. Snails were exposed for 3 hours in 45 mL of aerated solution in individual containers ($n = 6$ replicates per treatment). This duration was long enough to measure influx, yet short enough to avoid noticeable backflux of the radioisotope, as demonstrated in Crémazy et

al. (2019). After exposure, each snail was quickly rinsed in Ni-free test water, then for 5 min in 20 mL of 1 mM ethylenediaminetetraacetic (EDTA·2H₂O, ACS grade Fisher Scientific) prepared in Ni-free test water ([Ni] < 0.7 µg L⁻¹, method quantification limit), and finally rinsed quickly again in Ni-free test water. Soft tissues were dissected, weighed, then digested for 2 days at 65 °C in 50% HNO₃ v/v (ACS grade, Fisher Scientific). Filtered exposure water samples (0.45 µm polyethersulfone (PES) membrane, Membrane Solution, Dallas, TX, USA) were also collected at the beginning and at the end of the uptake test, for physicochemical analyses.

Chronic Ni toxicity tests

As shown in Table 3, three chronic toxicity experiments were conducted in the present study: Exp#2, Exp#3 and Exp#4. Exp#2 (conducted at UBC) was designed to evaluate water effects using UBC snails. In this experiment, 7-d SGR were measured with UBC snails exposed in UBC water or UM water. The test waters were spiked with 0, 10 and 250 µg L⁻¹ Ni and there were 5 replicates per treatment. The limited test duration and Ni treatments were due to the limited amount of shipped UM water.

Exp#3 (conducted at UM) was designed to further evaluate water effects (using UM snails) and to evaluate snail population effects in UM water. In this experiment, 7-d and 14-d SGR were measured with UM snails exposed in UBC or UM water and with UBC snails exposed in UM water. As in Exp#2, the test waters were spiked with 0, 10 and 250 µg L⁻¹ Ni and there were 5 replicates per treatment. Note that the same 7-d SGR measurements with UBC snails in UM water were performed at UBC (Exp#2) and UM (Exp#3), thereby facilitating assessment of test reproducibility between the two labs.

Finally, Exp#4 (conducted at UBC) was designed to further evaluate snail population effects (in UBC water). In this experiment, 14-d SGR were measured with UBC snails or UM snails exposed in UBC water. Five snails per treatment were exposed in bulk in 2 L of exposure water at 13 Ni concentrations ranging from 0 to 300 $\mu\text{g L}^{-1}$ Ni, as in Crémazy et al. (2018). The treatments were not replicated.

For all three tests, exposure solutions were prepared by spiking stock solutions of $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$, ACS grade, Fisher Scientific) into both waters, 24 h prior to test initiation. Snails ~2 weeks old (5 – 10 mg) were exposed in aerated exposure solutions, either individually (in 250 mL of water, Exp#2 and #3) or in bulk (five snails in 2 L of waters, Exp#4). Every day, water was renewed and snails were fed *ad libitum* with fresh washed romaine lettuce. Water samples were collected at day 1, 7 and 14 (when applicable), before water changes (24-h old water) and after water changes (fresh water), and filtered (0.45 μm PES membrane, Membrane Solution, Dallas, TX) for measurements of Ni concentration and other physicochemical parameters (pH, major cations and DOC concentrations). At the end of the test, snails were blotted dry and wet weight was determined with an accuracy of ± 0.1 mg with an analytical balance.

Physicochemical analyses and free Ni^{2+} measurements

The pH was measured with an Orion™ Green pH combination electrode (Fisher Scientific). Inorganic elements were analyzed by atomic absorption spectrometry (UBC: AA240 FS, Varian) in furnace mode (GFAAS) for dissolved Ni and in flame mode (FAAS) for dissolved Ca, Mg, K and Na. The DOC and DIC concentrations were measured with a Total Organic Carbon analyzer (V-series TOC analyzer, Shimadzu).

Instrument calibrations were verified with certified reference waters TMDA-54.5 and TM-25.4 from Natural Resources Canada (percent recovery was > 95%). A blank and calibration standard was re-analyzed every 8 (TOC analysis) or 12 (AAS analysis) samples to check and correct for potential instrument drift. For the short-term uptake tests, beta radiation from ^{63}Ni in water and snail tissues was measured as cpm (counts per minute) using an automated scintillation counter (LS 6500, Beckman Coulter), with appropriate quench correction (Crémazy et al. 2019).

To characterize potential differences in Ni bioavailability in UBC and UM waters, we measured Ni^{2+} concentrations in both waters using a solid-phase Ion Exchange Technique (IET). This technique has been shown to selectively measure Ni^{2+} species in natural waters (Worms and Wilkinson, 2008). These measurements were performed in the range of dissolved Ni concentrations eliciting growth inhibition in UBC water (10 – 220 $\mu\text{g L}^{-1}$). The working principle of IET has been described in details by Worms and Wilkinson (2008). Briefly, samples and standard solutions were passed through a column with 50 - 100 mg of cation-exchange resin (Dowex 50W-X8, 50-100 mesh; Sigma) until equilibrium was reached (*i.e.* when total dissolved [Ni] entering and exiting the resin column became equal). The resin was then eluted with nitric acid (Omni-trace grade, Sigma Aldrich) and the Ni concentration in the eluate was measured by GFAAS (Perkin Elmer PinAAcle 900T AA). After calibration with standard solutions of known Ni^{2+} concentrations, the Ni^{2+} concentration in the test solutions could be calculated from the Ni concentration in the eluates, for a given acid volume and resin mass.

Finally, to characterize potential differences in the Ni-binding properties of the dissolved organic matter (DOM) in UBC and UM waters, Fluorescence Excitation Emission Matrices (FEEMs) were measured using a spectrofluorometer (Varian ECLIPSE Cary fluorescence spectrophotometer (Varian Inc. Old Oak, NJ, USA)) and a 1-cm quartz cuvette (Starna Cells Inc.) (Tait et al. 2018). For scans, the excitation monochromator was varied from 200 to 450 nm in 10-nm increments, and for each excitation wavelength the sample emission was recorded at 250 to 600 nm in 1-nm increments. For fluorescence quenching experiments, Ni was titrated at pH 7.8 (pH adjusted using dilute NaOH additions). Samples were equilibrated 24 hours before measurement of the FEEM.

Experimental data analyses

If not stated otherwise, results are presented as mean $\pm$ standard error (SE) or as mean with 95% confidence intervals. Statistical analyses of the new experimental data were conducted with the software GraphPad Prism, with a significance level of 0.05.

The ion-exchange technique (IET) data in the two waters were analyzed by linear regression (IET-Ni²⁺ concentration as a function of total Ni concentration) and compared with an extra sum-of-squares F test.

Nickel internalization (uptake) rates (J_{int} , in $\mu\text{g}\cdot\text{g}^{-1}\cdot\text{h}^{-1}$) (Exp#1) were calculated using the following equation, with a = cpm g^{-1} of whole snail body (wet weight), b = cpm ml^{-1} in the water, c = the measured dissolved Ni concentration in the water ($\mu\text{g ml}^{-1}$) and t = time (3 h):

$$J_{int} = \frac{a \times c}{b \times t} \quad (\text{eq. 3})$$

The J_{int} values were compared at the different Ni exposure concentrations and for both water types, using a two-way ANOVA and Tukey's post hoc test.

Nickel chronic toxicity (Exp# 2 – 4) is reported both as effects on w_f (in mg wet weight) and effects on SGR (in d^{-1} , calculated with Eq. 1). For chronic toxicity tests conducted at three Ni concentrations (Exp# 2 and Exp#3), the growth measurements were compared at the various Ni exposure concentrations and for both water types, snail populations, or labs, using two-way ANOVAs and Tukey's post hoc tests. For the toxicity tests performed according to the Crémazy et al. (2018) protocol (Exp#4), data were fitted against Ni exposure concentrations with a three-parameter logistic equation after log transformation of dissolved Ni concentrations. The regressions in both waters were compared using an extra sum-of-squares F test.

RESULTS

Literature data

Table 2 shows the EC20s and EC50s obtained in the four literature studies, based on snail final biomass (includes final shell length) and SGR. The ECx estimates based on biomass were consistently lower than the estimates based on SGR, with the exception of Nys et al. (2016) where they were similar. Estimated $EC20_{biomass}$ were up to 27-fold (average 5.7-fold) lower than $EC20_{SGR}$ and estimated $EC50_{biomass}$ were up to >11-fold (average 3.3-fold) lower than $EC50_{SGR}$. Inter-study variability was higher when toxicity was expressed based on biomass compared to SGR. Estimated $EC20_{biomass}$ and $EC50_{biomass}$ observed in the various tests varied by >126-fold and 109-fold respectively, with the greatest difference observed between the Niyogi et al. (2014) and the Crémazy et

al. (2018) studies. In comparison, the inter-study variation was about 20-fold and 22-fold when considering $EC_{20_{SGR}}$ ($7.3 - 150 \mu g L^{-1}$) and $EC_{50_{SGR}}$ ($10 - 220 \mu g L^{-1}$) respectively, with the greatest difference observed between the Schlekat et al. (2010) (Calapooia River water) and the Crémazy et al. (2018) studies.

Table 2 also reports $EC_{50_{biomass}}$ and $EC_{20_{SGR}}$ predicted with the chronic Ni BLM for *L. stagnalis* developed by Schlekat et al. (2010) and Nys et al. (2016), respectively. The performance of each BLM for the prediction of the literature ECx data is illustrated by Figure 1. The biomass BLM explained the 109-fold variation in $EC_{50_{biomass}}$ within a 23-fold maximum deviation from prediction. Notably, Ni toxicity was over-predicted by a factor of 23 in Crémazy et al. (2018) and under-predicted by a factor of 5 in Niyogi et al. (2014) (Table 2, Figure 1a). The SGR BLM explained the 22-fold inter-study variability within a factor of 6.6 (Table 2, Figure 1b). It over-predicted most measured $EC_{20_{SGR}}$ values, with the exception of the Crémazy et al. (2018) $EC_{20_{SGR}}$ that was under-predicted by a factor 3.6. Because of the very similar water chemistry in Niyogi et al. (2014) (UM water) and Crémazy et al. (2018) (UBC water), both BLMs predicted nearly identical chronic Ni toxicity in these two studies (Table 2, Figure 1). Yet, Crémazy et al. (2018) measured $EC_{50_{biomass}}$ and $EC_{20_{SGR}}$ that were respectively 109 times and 16 times higher than in Niyogi et al. (2014).

In UBC and UM waters respectively, Crémazy et al. (2019) and Niyogi et al. (2014) measured short-term Ni uptake flux rates in juvenile *L. stagnalis* using ^{63}Ni . The comparison of these literature data is shown in Figure SI.1 (Supplemental Information),

where J_{int} values in UM water (Niyogi et al., 2014) were about 5-fold higher than in UBC water at comparable dissolved Ni concentrations (Crémazy et al. 2019).

Nickel speciation in test waters

At the observed $\text{EC}_{20\text{SGR}}$ in UBC water ($150 \mu\text{g L}^{-1}$) and UM water ($9.4 \mu\text{g L}^{-1}$), WHAM VII predicted that 71% and 51% of Ni was present as free bioavailable Ni^{2+} respectively. At the observed $\text{EC}_{50\text{biomass}}$ ($160 \mu\text{g L}^{-1}$ in UBC water and $1.5 \mu\text{g L}^{-1}$ in UM water), these predictions were respectively 65% and 35%. The 6-fold SO_4 concentration difference in the two waters had only a small effect on these Ni^{2+} percentages, since Ni- SO_4 complexes were not very abundant (6% and 1% of total Ni species in UBC and UM waters respectively). The lower proportion of Ni^{2+} in UM water was mainly attributed to its 3-fold higher DOC concentration, leading to increased Ni-DOC complexation.

Figure 2 shows the $[\text{Ni}^{2+}]$ measured by IET as a function of the total $[\text{Ni}]$ in each water. These measurements seemed to slightly disagree with the WHAM predictions, since the measured $\%\text{Ni}^{2+}$ was $58 \pm 5 \%$ in UBC water and $76 \pm 6\%$ in UM water. However, our analysis of the IET data concluded that there was no statistical difference in Ni speciation between the two waters (F-test, $p = 0.1389$; regression equation: $\log [\text{Ni}^{2+}] = 0.7998 \times \log [\text{Ni}] - 7.379$).

Fluorescence Excitation Emission Matrices (FEEM) of UBC and UM waters indicated that the DOCs of both waters exhibit fulvic-like components, which could be detected in the emission – excitation ranges of 250–390 nm to 460–520 nm (Figure 3). However, DOM in the UBC water also has a tyrosine-like component, which could be

detected in the emission – excitation ranges of 225–270 nm to 300 nm. When UBC water was incubated with 40 $\mu\text{g L}^{-1}$ of total Ni, fluorescence of the tyrosine-like peak increased (Fig. 3c), likely indicating interactions between this fluorophore and nickel ions.

Short-term Ni uptake

A slightly significant effect of water type was observed in newly measured short-term Ni uptake flux rates in Exp#1 (two-way ANOVA, $p = 0.039$, 4% of total variation) (Figure 4). However, the Tukey's test did not find any difference between the water types for each of the tested [Ni]. Indeed, the 3-h Ni uptake flux rates measured in UBC snails were very similar in UBC water and in UM water, at the two tested Ni concentrations. These Ni flux rate levels were similar to the levels reported by Crémazy et al. (2019) for UBC snails in UBC water.

Chronic Ni toxicity

No mortality was observed among the control snails across all the toxicity tests performed in this study. Average SGR in the control snails was $0.221 \pm 0.008 \text{ d}^{-1}$ ($n = 42$) and ranged from 0.098 to 0.314 d^{-1} . These growth rates are consistent with control SGR measured under similar test conditions by Schlekot et al. (2010), Niyogi et al. (2014) and Crémazy et al. (2018): from 0.11 to 0.24 d^{-1} with an average of 0.20 d^{-1} . Note that Nys et al. (2016) measured much lower control SGR, from 0.022 to 0.047 d^{-1} (average of 0.032 d^{-1}), probably due to differences in test conditions (darker light cycle, lower temperature and less nutritious food). No conclusion on test validity was made based on the growth

rate of the control snails, due to lack of information on what is considered acceptable under our test conditions.

Figure 5 shows snail growth, measured as either specific growth rate (SGR, left panel) or biomass (w_f , right panel) as a function of dissolved Ni concentration, in Exp#2 and Exp#3, which tested the effects of test location (lab), water type and snail population on chronic Ni toxicity. Figure 6 also shows effects of snail population source on chronic Ni toxicity, but in Exp#4 performed with a different experimental design.

Inter-lab experimental reproducibility (lab effects) was evaluated with an identical 7-d chronic toxicity test performed at UBC (Exp#2) and at UM (Exp#3) (Fig. 5a and a'). Snails at UBC grew consistently less than at UM, at each Ni treatment (including the controls). Significant decreases of snail growth were observed at $250 \mu\text{g L}^{-1}$ Ni in both labs. When considering SGR (Fig. 5a), this growth inhibition at $250 \mu\text{g L}^{-1}$ was 3-fold higher in UBC compared to UM, but no significant interaction between test location and Ni toxicity was observed. When considering biomass (w_f) (Fig. 5a'), growth inhibition at $250 \mu\text{g L}^{-1}$ was only significant at UM, although both w_f decreased by the same 3-fold factor in both labs. Overall, these data suggest that, although test location affected absolute snail growth, it had little impact on chronic Ni toxicity.

Effects of water type were evaluated with UBC snails (Fig. 5b and b', Exp# 2) and with UM snails (Fig. 5c and c', Exp#3). As in the above test, effects of Ni on snail growth were only observed at $250 \mu\text{g L}^{-1}$ Ni. There was no effect of water type on snail growth. A relatively small interaction effect (10% of total variability) was observed with

SGR of UBC snails (Fig. 5b), such that growth tended to be higher in UM water at low Ni concentrations, but lower in UM water at high Ni concentration.

Effects of snail population source were evaluated in UM water (Figure 5d and d', Exp#3) and in UBC water (Figure 6a and a', Exp#4). In UM water (Exp#3), both main effects (Ni concentration and population) were significant, but that there was also an important, significant interaction effect (Figure 5d and d'). Effects of Ni on snail growth only occurred at 250 $\mu\text{g L}^{-1}$ Ni, with UBC snails growing significantly less than UM snails at the highest Ni concentration (Fig. 5d and d'). However, this finding was not confirmed in UBC water (Exp#4) (Fig. 6a and a'). In this latter test, Ni effects also occurred in the 100 – 300 $\mu\text{g L}^{-1}$ Ni range and different concentration-response curves (F-test, $p < 0.001$) also confirmed a snail population effect on Ni toxicity. However, this time UM snails were more sensitive to Ni. Indeed, estimated $\text{EC}_{50\text{SGR}}$ was 230 [190 - 270] $\mu\text{g L}^{-1}$ for UBC snails and of 150 [130 – 170] $\mu\text{g L}^{-1}$ for UM snails based on SGR (Fig. 6a). Similarly, estimated $\text{EC}_{50\text{biomass}}$ was 180 [140 - 240] $\mu\text{g L}^{-1}$ for UBC snails and of 110 [96 – 130] $\mu\text{g L}^{-1}$ for UM snails (Fig. 6a') (see Table SI.2 in the Supporting Information for detailed regression results). Note however than when considering $\text{EC}_{20\text{s}}$, there was no significant difference between the two snail populations, mainly due to larger 95% CL on these estimates. For UBC snails, $\text{EC}_{20\text{SGR}}$ was 140 [100 - 190] $\mu\text{g L}^{-1}$ and $\text{EC}_{20\text{biomass}}$ was 110 [63 - 190] $\mu\text{g L}^{-1}$. For UM snails, $\text{EC}_{20\text{SGR}}$ was 100 [82 – 130] $\mu\text{g L}^{-1}$ and $\text{EC}_{20\text{biomass}}$ was 82 [61 – 110] $\mu\text{g L}^{-1}$.

DISCUSSION

Major findings

First, we showed that a large proportion of the variability in published Ni toxicity data could be accounted for by bioavailability correction (using the BLM) and by using specific growth rate (SGR) rather than biomass (w_f) as a response variable. The latter is mainly because SGR is less dependent on test duration, which varied by 2-fold among the studies. Second, our new experiments showed that test water composition (UBC vs. UM water) or test location (at UBC vs. UM) had only small effects or no effects on the short-term uptake and chronic toxicity of Ni. We observed snail population source (UBC vs. UM snails) effects on Ni toxicity, but these effects were inconsistent and small relative to the differences between Niyogi et al. (2014) and Crémazy et al. (2018). Finally, the Niyogi et al. (2014) toxicity data could not be reproduced, as snail Ni sensitivity measured in the present study was systematically closer to the sensitivity reported in Crémazy et al. (2018) (*i.e.* with toxic effects in the low hundreds $\mu\text{g L}^{-1}$).

*Variability in chronic Ni toxicity to *L. stagnalis* in the literature*

Both literature EC20s and EC50s are reported in the present study. The low-effect estimate EC20 is typically considered a better parameter than the EC50 for regulatory purposes. However, among all the statistical estimates, the EC50 generally has the narrowest confidence interval (*i.e.* the lowest measurement uncertainties), making it a more robust parameter for inter-study comparison (Christensen et al. 2009). Furthermore, we reported ECx based on two response variables: final biomass (used in Schlekot et al.

(2010) and Niyogi et al. (2014)) and SGR (used in Nys et al. (2016) and Crémazy et al. (2018)). Although both response variables are derived from the same measurements and assess percent growth inhibition (by relating to the response in control treatments), the resulting ECx figures are markedly different, for systemic and mathematical reasons. Indeed, for organisms with exponential growth (*e.g.* *Lymnaea* snails, most unicellular algae and some aquatic plants), small changes in growth rate result in much larger changes in biomass. Thus, biomass will generally give lower ECx estimates than SGR. This is indeed what was generally observed across the various literature studies (Table 2). The use of biomass or growth rate to assess growth inhibition is an old controversy that has received much attention in algae studies (Nyholm 1985; Ratte et al. 1998; Bergtold and Dohmen 2011) but remains globally unresolved in the regulatory community (Weyers and Vollmer 2000; OECD 2011). On one hand, it is common practice for regulatory purposes to use the response variable exhibiting the lowest value. On the other hand, for the same reasons biomass is mathematically more sensitive than SGR, it also has a larger variance (Ratte et al. 1998). Indeed, SGR averages over the growth curve, whereas final biomass sums up all deviations in the growth and thus also the variability. As a result of its higher variance, biomass is not always found to be more sensitive for detecting biological effects than SGR, as observed in some tests of the present study (Fig. 5b vs. 5 b' and Fig. 5c vs. 5c'). However, had these tests been conducted for a longer time (*i.e.* > 7 d), it is likely that biomass would have been found more sensitive than SGR (ECx_{biomass} estimates substantially decrease with time). Indeed, relative to SGR, biomass is more dependent on test conditions, such as biomass at the start of the test, snail intrinsic growth rate, and more importantly test duration (Nyholm 1985; Ratte et al. 1998;

Bergtold and Dohmen 2011). Thus, the use of biomass has been argued to be one of the reasons for the lack of comparability among algal inhibition studies (Nyholm 1985). The larger variance of the toxicity data derived from biomass could possibly be diminished by strict standardization of test conditions across studies, although this may not always be practical. In part due to the absence of a standardized method for testing growth effects with *L. stagnalis*, the four literature studies had many methodological differences (*e.g.* test duration, starting size/age, photoperiod, temperature, food) (Table 1). Based on test duration differences alone, these tests are best compared based on toxicity estimates derived from SGR, which are much less time-dependent (Nyholm 1985; Ratte et al. 1998; Bergtold and Dohmen 2011). Indeed, in the present study, test duration (7 or 14 d) had no or negligible effects on SGR inhibition by Ni (Fig. SI.2 in the Supplemental Information). Likewise, time (7, 14 and 21 d) had no effect on SGR inhibition in Niyogi et al. (2014) (Fig. SI.2). It was therefore not surprising that we observed a 5- to 6-fold lower inter-study variability when comparing ECx_{SGR} rather than $ECx_{biomass}$ (Figure 1 and Table 2).

Even with SGR, there was still a ~20-fold variation in chronic Ni toxicity among the various literature tests. Beyond differences in the test methods, intra-species metal toxicity variability is usually largely attributed to differences in test water chemistry (*e.g.* pH, DOC concentration, major cation concentrations, etc.) affecting metal bioavailability to aquatic organisms (Paquin et al. 2002). According to the BLM conceptual framework, metal bioavailability (and thus toxicity) is associated with the free metal ion concentration (*e.g.* Ni^{2+}). Major cations (*e.g.* Ca^{2+} and Mg^{2+}) decrease metal bioavailability by competing with the free metal ion for binding to biologically sensitive

sites (biotic ligands, *e.g.* membrane transporters), while abiotic ligands (*e.g.* DOC and Cl^-) compete with these biotic ligands for binding the free metal ions. Thus, both major cations and abiotic ligands exert protection against metal toxicity. The four Ni literature studies were performed in test waters of varying water chemistry ($\text{pH} = [6.9 - 8.6]$, $[\text{Ca}^{2+}] = [0.085 - 1.80 \text{ mM}]$, $[\text{Mg}^{2+}] = [0.033 - 0.699 \text{ mM}]$, $\text{DOC} = [0 - 7.1 \text{ mg L}^{-1}]$). Water chemistry effects can be accounted for by the use of computational BLMs that have been parameterized for a given metal, organism and endpoint. Schlekot et al. (2010) and Nys et al. (2016) adjusted the species-sensitivity parameter (to very different values) of an existing *Ceriodaphnia dubia* chronic Ni BLM (eq. 2) in order to fit their $\text{EC}_{50_{\text{biomass}}}$ and $\text{EC}_{20_{\text{SGR}}}$ measurements respectively with *L. stagnalis*. The biomass BLM explained the 109-fold variation in $\text{EC}_{50_{\text{biomass}}}$ within a 23-fold maximum deviation from prediction. However, as discussed in the earlier paragraph, biomass should not be used when comparing toxicity tests with different test durations. The SGR BLM explained the 22-fold variation in $\text{EC}_{20_{\text{SGR}}}$ within a factor of 6.6-fold.

The residual inter-study variability after bioavailability-correction could reflect uncertainties on water chemistry (*e.g.* pH) measurements that are fed into the BLM, and/or may indicate that the BLM is not adequately parameterized. Regarding this latter point, it is important to remember that the Ni BLMs used here were developed for *C. dubia* and only their sensitivity parameters were adjusted to fit very small *L. stagnalis* datasets (respectively a 5-point and 1-point dataset in Schlekot et al. (2010) and in Nys et al. (2016)). BLM predictions could probably be improved if a new model was developed from a dedicated and larger *L. stagnalis* toxicity dataset. However, it is very unlikely that

the BLM could be improved to the point of reconciling the Niyogi et al. (2014) and Crémazy et al. (2018) toxicity data within an acceptable margin of error. Indeed, the current BLM modeling exercise highlighted the surprising 16-fold difference in the $EC_{20_{SGR}}$ measured by Niyogi et al. (2014) and Crémazy et al. (2018). For these two studies, the chemistry of the test waters (UM and UBC waters) was quite similar (Table 4), so that the BLMs predicted nearly identical Ni toxicity. More precisely, at the $EC_{20_{SGR}}$, WHAM VII predicted slightly more bioavailable Ni^{2+} in UBC water, because of its 3-fold lower DOC concentration. However, this lower complexation-based protection in UBC was counteracted by a larger competition-based protection, due to the 2-fold higher $[Ca^{2+}]$ in UBC water. Yet, despite a similar Ni bioavailability predicted in both studies, the chronic Ni toxicity (Figure 1) and uptake rate (Figure SI.1) were much higher in Niyogi et al. (2014) at UM than in Crémazy et al. (2018 and 2019) at UBC, respectively. These observations suggested that Ni bioavailability predictions were somehow erroneous in these studies, which triggered additional tests.

Effects of test waters

With new measurements, we first investigated if the reported lower Ni uptake rates and lower chronic toxicity in UBC water could be due to lower Ni^{2+} concentrations, in contradiction with WHAM VII predictions. This lower Ni bioavailability could be due to the presence of an uncharacterized Ni-complexing ligand, or to a DOC in UBC water with remarkably larger Ni-binding properties. Our FEEM results suggested that despite the lower concentration of DOC in UBC water, an unknown tyrosine-like component with unusual Ni-binding properties might be reducing the concentration of free ionic Ni^{2+} .

below that in UM water at the same total Ni concentration. However, the IET analysis indicated that overall, free ionic Ni^{2+} was not statistically different in UBC and UM water. The similar Ni bioavailability in the two waters was further supported by new measurements of Ni uptake rates and chronic toxicity, which showed no significant difference between snails exposed to the two test waters. In these tests, the Ni uptake rates and chronic Ni toxicity were similar to those observed in Crémazy et al. (2019) and (2018) respectively. Altogether, these different measurements suggest that, as expected from their similar chemical compositions, difference in water chemistry could not explain the large difference in Ni uptake rate and chronic toxicity observed between Niyogi et al. (2014) and Crémazy et al. (2018, 2019).

Snail population and lab effects

We then investigated if UBC snails were inherently more tolerant to Ni than the UM snails. Since these two snail cultures had been separated for over a decade, this was a distinct possibility. Over this time, the two isolated and small-sized cultures may have experienced significant genetic differentiation (and various levels of genetic impoverishment compared to wild animals), which can lead to different sensitivities to contaminants (Nowak et al. 2008). Two new toxicity tests concluded that there was a relatively small difference in Ni sensitivity between the snail cultures: a 2.3-fold difference in SGR measured at $250 \mu\text{g}\cdot\text{L}^{-1}$ Ni in UM water (Fig. 5d) and a 1.5-fold difference in $\text{EC}_{50\text{SGR}}$ measured in UBC water (Fig. 6a). However, in the former experiment, UBC snails exhibited a higher Ni sensitivity than the UM snails (Fig. 5d), while the opposite was found in the latter experiment (Fig. 6a). While the UBC snails

showed a Ni sensitivity similar to that reported in Crémazy et al. (2018), the UM snails appeared much less sensitive in these new tests than in Niyogi et al. (2014). In fact, using UM snails in either test water, the present study could not reproduce the extremely low chronic Ni toxicity levels ($>$ one order of magnitude lower) reported by Niyogi et al. (2014). Using the same water, snail source population and methods, we showed that between-lab reproducibility for the assessment of chronic Ni toxicity was quite reasonable (Fig. 5a and 5a'). Thus, potential explanations for these discrepancies are that i) the UM snail culture has changed since the Niyogi et al. (2014) study and/or that ii) there were important differences in test conditions and methods. Because there was no apparent reason that could explain the former point, we focus on the latter in the following discussion.

Other potentially important factors

Beside water composition and snail culture, there were differences of methodological nature between the Ni studies, *e.g.*, test animal housing, exposure duration, test food, and snail age/weight at test start (Table 1).

Snails in Schlekot et al. (2010) and Niyogi et al. (2014) (with the highest Ni sensitivity, Table 2 and Fig. 1) were exposed in individual containers, while snails in Crémazy et al. (2018) and Nys et al. (2016) (with the lowest Ni sensitivity) were exposed in groups. Multiple snail studies have shown that individual interaction through waterborne substances (pheromones and metabolic chemicals) may affect the growth of conspecifics, in both positive and negative manners (Levy et al. 1973; Thomas and Aram 1974; Thomas and Benjamin 1974; Thomas et al. 1975). However, in the present study,

there was no remarkable difference in Ni sensitivity between snails exposed in group (Fig. 6a) or in isolation (Fig. 5a, 5b, 5c, 5d).

Test duration varied from 14 to 30 days in the literature studies. Although this factor will strongly affect toxicity estimates based on biomass, ECxs derived from SGR are not as dependent on test duration (Bergtold and Dohmen 2011). Indeed, in the present study and in Niyogi et al. (2014), test duration had little or no effects on SGR inhibition (Fig. SI.2 in the Supplemental Information), indicating no important change in snail sensitivity throughout the test.

Growth being directly related to energetic resources, food quality may greatly affect this biological response in toxicity tests. Great pond snails raised on different diets have shown significant variation in their chemical sensitivity (Fidder et al. 2018). For example, Reategui-Zirena et al. (2016) showed that *L. stagnalis* growth was more inhibited by Cd when snails were fed high-caloric pellets rather than romaine lettuce (although their control growth rate was higher). These food effects have also been shown to be transgenerational in other snails (Plautz et al. 2013), so that parental diet should also be taken in consideration. Although *L. stagnalis* feeds on plants, carrion, algae and microorganisms in their natural environment (Fidder et al. 2018), a typical laboratory diet for *L. stagnalis* is romaine lettuce. During the toxicity tests, Crémazy et al. (2018) and Niyogi et al. (2014) used romaine lettuce, Nys et al. (2016) used butterhead lettuce and Schlekot et al. (2010) used a mix of carrot, sweet potato and romaine lettuce (c.f. Table 1). Also, only the lettuce in Nys et al. (2016) was of organic quality. Furthermore, newly-hatched snails in Crémazy et al. (2018) and Niyogi et al. (2014) were fed with sweet

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potato prior to the toxicity test. These variations in food quality may significantly influence the bioenergetic resources and tolerance of snails to Ni. However, because the snails in Crémazy et al. (2018) and Niyogi et al. (2014) were raised and tested using the same diet, food does not appear to be an important factor to explain the large variation between the two studies. Yet, it is possible that the lettuces used in these two studies had different nutrient contents. Indeed, Hartz et al. (2007) showed that the nutrient composition (*e.g.*, Ca and Mg) of romaine lettuce can vary substantially with its source. Thus, we believe that this particular factor should be further investigated.

Age/weight of snails at the beginning of toxicity tests was different in each literature study, ranging from < 24 hours old (Schlekat et al. 2010) to 3 weeks old (Crémazy et al. 2018). With a life expectancy around 2 years for *L. stagnalis* (Amorim et al. 2019), a 3-week age difference may appear insignificant. Yet, critical physiological development occurs at a fast rate during early life (embryonic and juvenile) stages, and surface area-to volume ratio decreases progressively. Thus, newly-hatched snails may be significantly more impacted by Ni than 3-week old snails, because of lower metabolic capacity for excretion or greater relative surface area for uptake, and/or enhanced sensitivity (*e.g.* due to functional immaturity of detoxification processes). The age difference between the snails used in Niyogi et al. (2014), Nys et al. (2016) and Schlekat et al. (2010) versus Crémazy et al. (2018) and the present study was relatively small (2 to 3 weeks), with the age ranges over the toxicity tests overlapping among studies. Yet, a sensitive early life stage, even if brief, may result in a prolonged lower growth trajectory. The most sensitive reports (Schlekat et al. 2010; Niyogi et al. 2014; Nys et al. 2016) all

started with younger animals (<8 days at test onset) than the Crémazy et al. (2018) and current study (>14 days at test onset). In the four published studies, we found no statistical correlation ($p = 0.1211$) between snail age at the start of toxicity testing and chronic Ni toxicity (Fig. SI.3). For this analysis, we compared snail age with the BLM EC20_{SGR} residuals (differences between observed and predicted EC20_{SGR} in Table 2), to account for the bioavailability effects predicted by the BLM. However, this analysis was made on a relatively small dataset, with potential confounding factors (such as the factors discussed above) that may have masked the age effect. The possible influence of snail age should certainly be further investigated.

CONCLUSIONS AND PERSPECTIVES

While the Crémazy et al. (2018) study appears to be an outlier among the various literature studies, its relatively high ECs were reproduced multiple times in the present study, and as noted earlier, it was an outlier only for Ni, and not other metals. Conversely, while the Niyogi et al. (2014) data could not be reproduced, the very low ECs of that study were supported by other studies (Schlekat et al. 2010; Peters et al. 2014; Nys et al. 2016). Our present investigation does not conclude that one study has better assessed the “intrinsic sensitivity” of the great pond snail to Ni. It simply emphasizes that this sensitivity varies greatly under the influence of abiotic and/or biotic factors that remain to be characterized. Differences in metal bioavailability or source of snails could not explain the large Ni toxicity difference between Niyogi et al. (2014) and Crémazy et al. (2018). In future studies, we recommend characterizing separately the possible influence(s) of test animal housing, test duration, diet and snail age as possible

toxicity modifying factors. Taken together, these various parameters may have a significant impact on toxicity test results. Until this variability is resolved/better understood, caution should be used before applying *L. stagnalis* data in a regulatory setting or in a guideline. Our study emphasizes the importance of developing standard methodologies for toxicity testing, especially for highly sensitivity species whose toxicity levels may affect regulatory guidelines. We also encourage further studies aimed at unraveling the exact mechanism leading to snail growth inhibition during chronic Ni exposure. This knowledge will be valuable toward the development of a standardized toxicity test for assessing growth effects with *L. stagnalis*. Ultimately, such knowledge and methodological guidelines will help better assess Ni environmental risk to this important freshwater species.

REFERENCES

- Amorim J, Abreu I, Rodrigues P, Peixoto D, Pinheiro C, Saraiva A, Carvalho AP, Guimarães L, Oliva-Teles L. 2019. *Lymnaea stagnalis* as a freshwater model invertebrate for ecotoxicological studies. Science of The Total Environment. 669:11–28. doi:10.1016/j.scitotenv.2019.03.035.
- Bergtold M, Dohmen GP. 2011. Biomass or growth rate endpoint for algae and aquatic plants: Relevance for the aquatic risk assessment of herbicides. Integrated Environmental Assessment and Management. 7(2):237–247. doi:10.1002/ieam.136.

Brancho J. 2017. Review of regulatory policies for copper and silver water quality criteria [Honors Thesis]. Kent State University Honors College.

Brix KV, Esbaugh AJ, Grosell M. 2011. The toxicity and physiological effects of copper on the freshwater pulmonate snail, *Lymnaea stagnalis*. *Comparative Biochemistry and Physiology Part C: Toxicology & Pharmacology*. 154(3):261–267. doi:10.1016/j.cbpc.2011.06.004.

Brix KV, Esbaugh AJ, Munley KM, Grosell M. 2012. Investigations into the mechanism of lead toxicity to the freshwater pulmonate snail, *Lymnaea stagnalis*. *Aquatic Toxicology*. 106–107:147–156. doi:10.1016/j.aquatox.2011.11.007.

Brix KV, Schlekot CE, Garman ER. 2017. The mechanisms of nickel toxicity in aquatic environments: An adverse outcome pathway analysis. *Environmental Toxicology and Chemistry*. 36:128–1137.

Bronmark C. 1994. Effects of Tench and Perch on Interactions in a Freshwater, Benthic Food Chain. *Ecology*. 75(6):1818–1828. doi:10.2307/1939640.

Christensen ER, Kusk KO, Nyholm N. 2009. Dose–response regressions for algal growth and similar continuous endpoints: Calculation of effective concentrations. *Environmental Toxicology and Chemistry*. 28(4):826–835. doi:10.1897/08-068R.1.

- Crémazy A, Brix KV, Wood CM. 2018. Chronic toxicity of binary mixtures of six metals (Ag, Cd, Cu, Ni, Pb, and Zn) to the great pond snail *Lymnaea stagnalis*. *Environmental Science & Technology*. 52(10):5979–5988. doi:10.1021/acs.est.7b06554.
- Crémazy A, Brix KV, Wood CM. 2019. Using the Biotic Ligand Model framework to investigate binary metal interactions on the uptake of Ag, Cd, Cu, Ni, Pb and Zn in the freshwater snail *Lymnaea stagnalis*. *Science of The Total Environment*. 647:1611–1625. doi:10.1016/j.scitotenv.2018.07.455.
- De Schamphelaere KAC, Janssen CR. 2010. Cross-phylum extrapolation of the *Daphnia magna* chronic biotic ligand model for zinc to the snail *Lymnaea stagnalis* and the rotifer *Brachionus calyciflorus*. *Science of The Total Environment*. 408(22):5414–5422. doi:10.1016/j.scitotenv.2010.07.043.
- De Schamphelaere KAC, Koene JM, Heijerick DG, Janssen CR. 2008. Reduction of growth and haemolymph Ca levels in the freshwater snail *Lymnaea stagnalis* chronically exposed to cobalt. *Ecotoxicology and Environmental Safety*. 71(1):65–70. doi:10.1016/j.ecoenv.2007.07.004.
- European Chemicals Bureau. 2008. European Union Risk Assessment Report: nickel. Ispra, Italy: European Commission, Joint Research Centre, European Chemicals Bureau.

- Fidder BN, Reátegui-Zirena EG, Salice CJ. 2018. Diet quality affects chemical tolerance in the freshwater snail *Lymnaea stagnalis*. *Environmental Toxicology and Chemistry*. 37(4):1158–1167. doi:10.1002/etc.4064.
- Grosell M, Brix KV. 2009. High net calcium uptake explains the hypersensitivity of the freshwater pulmonate snail, *Lymnaea stagnalis*, to chronic lead exposure. *Aquatic Toxicology*. 91(4):302–311. doi:10.1016/j.aquatox.2008.10.012.
- Grosell M, Gerdes RM, Brix KV. 2006. Chronic toxicity of lead to three freshwater invertebrates—*Brachionus calyciflorus*, *Chironomus tentans*, and *Lymnaea stagnalis*. *Environmental Toxicology and Chemistry*. 25(1):97–104. doi:10.1897/04-654R.1.
- Hartz TK, Johnstone PR, Williams E, Smith RF. 2007. Establishing lettuce leaf nutrient optimum ranges through dris analysis. *HortScience*. 42:143–146.
- Hommen U, Knopf B, Rüdell H, Schäfers C, Schamphelaere KD, Schlekert C, Garman ER. 2016. A microcosm study to support aquatic risk assessment of nickel: Community-level effects and comparison with bioavailability-normalized species sensitivity distributions. *Environmental Toxicology and Chemistry*. 35(5):1172–1182. doi:10.1002/etc.3255.
- Leonard EM, Wood CM. 2013. Acute toxicity, critical body residues, Michaelis–Menten analysis of bioaccumulation, and ionoregulatory disturbance in response to waterborne nickel in four invertebrates: *Chironomus riparius*, *Lymnaea stagnalis*,

Lumbriculus variegatus and *Daphnia pulex*. Comparative Biochemistry and Physiology Part C: Toxicology & Pharmacology. 158(1):10–21. doi:10.1016/j.cbpc.2013.03.008.

Levy MG, Tunis M, Isseroff H. 1973. Population control in snails by natural inhibitors. Nature. 241(5384):65–66. doi:10.1038/241065a0.

Niyogi S, Brix KV, Grosell M. 2014. Effects of chronic waterborne nickel exposure on growth, ion homeostasis, acid-base balance, and nickel uptake in the freshwater pulmonate snail, *Lymnaea stagnalis*. Aquatic Toxicology. 150:36–44. doi:10.1016/j.aquatox.2014.02.012.

Nowak C, Czeikowitz A, Vogt C, Oetken M, Streit B, Schwenk K. 2008. Variation in sensitivity to cadmium among genetically characterized laboratory strains of the midge *Chironomus riparius*. Chemosphere. 71(10):1950–1956. doi:10.1016/j.chemosphere.2007.12.023.

Nyholm N. 1985. Response variable in algal growth inhibition tests—Biomass or growth rate? Water Research. 19(3):273–279. doi:10.1016/0043-1354(85)90085-5.

Nys C, Janssen CR, Sprang PV, Schamphelaere KACD. 2016. The effect of pH on chronic aquatic nickel toxicity is dependent on the pH itself: Extending the chronic nickel bioavailability models. Environmental Toxicology and Chemistry. 35(5):1097–1106. doi:10.1002/etc.3232.

- Nyström P, Pérez JR. 1998. Crayfish predation on the common pond snail (*Lymnaea stagnalis*): the effect of habitat complexity and snail size on foraging efficiency. *Hydrobiologia*. 368(1):201–208. doi:10.1023/A:1003266603371.
- OECD. 2011. Test No. 201: Freshwater Alga and Cyanobacteria, Growth Inhibition Test. OECD Publishing OECD Guidelines for the Testing of Chemicals. [accessed 2020 Feb 22]. https://read.oecd-ilibrary.org/environment/test-no-201-alga-growth-inhibition-test_9789264069923-en#page1.
- Pais N. 2012. Studies on waterborne cadmium exposure to *lymnaea stagnalis* in varying water qualities and the development of a novel tissue residue approach [Master of Science Thesis]. Wilfrid Laurier University. <https://scholars.wlu.ca/etd/837>.
- Paquin PR, Gorsuch JW, Apte S, Batley GE, Bowles KC, Campbell PGC, Delos CG, Di Toro DM, Dwyer RL, Galvez F, et al. 2002. The biotic ligand model: a historical overview. *Comparative Biochemistry and Physiology Part C: Toxicology & Pharmacology*. 133(1–2):3–35. doi:10.1016/S1532-0456(02)00112-6.
- Peters A, Simpson P, Merrington G, Schlekot C, Rogevich-Garman E. 2014. Assessment of the effects of nickel on benthic macroinvertebrates in the field. *Environ Sci Pollut Res*. 21(1):193–204. doi:10.1007/s11356-013-1851-2.
- Plautz SC, Funkhouser MA, Salice CJ. 2013. New insights into parental effects and toxicity: Mate availability and diet in the parental environment affect offspring

responses to contaminants. *Environmental Pollution*. 180:41–47.
doi:10.1016/j.envpol.2013.05.009.

Ratte HT, Hammers-Wirtz M, Cleuvers M. 1998. Influence of the growth pattern on the EC50 of Cell Number, Biomass Integral and Growth Rate in the Algae Growth Inhibition Test | Umweltbundesamt. Aachen, Germany: RWTH Aachen University of Technology Umweltbundesamt Report No.: Project No. 360 030 10. [accessed 2020 Feb 22]. <https://www.umweltbundesamt.de/en/publikationen/influence-of-growth-pattern-on-ec50-of-cell-number>.

Reátegui-Zirena EG, Fidler BN, Salice CJ. 2016. A cost or a benefit? Counterintuitive effects of diet quality and cadmium in *Lymnaea stagnalis*. *Ecotoxicology*. 25(10):1771–1781. doi:10.1007/s10646-016-1720-0.

Schlekat CE, Van Genderen E, De Schamphelaere KAC, Antunes PMC, Rogevich EC, Stubblefield WA. 2010. Cross-species extrapolation of chronic nickel Biotic Ligand Models. *Science of The Total Environment*. 408(24):6148–6157. doi:10.1016/j.scitotenv.2010.09.012.

Tait TN, McGeer JC, Smith DS. 2018. Testing the underlying chemical principles of the biotic ligand model (BLM) to marine copper systems: measuring copper speciation using fluorescence quenching. *Bull Environ Contam Toxicol*. 100(1):76–81. doi:10.1007/s00128-017-2262-8.

- Thomas JD, Aram RH. 1974. The chemical ecology of *biomphalaria glabrata* (say): The effects of media homotypically conditioned by adult snails on the growth of juveniles. *Journal of Experimental Zoology*. 190(3):329–339. doi:10.1002/jez.1401900309.
- Thomas JD, Benjamin M. 1974. The effects of population density on growth and reproduction of *biomphalaria glabrata* (say) (gasteropoda: pulmonata). *Journal of Animal Ecology*. 43(1):31–50. doi:10.2307/3156.
- Thomas JD, Goldsworthy GJ, Aram RH. 1975. Studies on the chemical ecology of snails: the effect of chemical conditioning by adult snails on the growth of juvenile snails. *Journal of Animal Ecology*. 44(1):1–27. doi:10.2307/3849.
- Tipping E, Loft S, Sonke JE. 2011. Humic Ion-Binding Model VII: a revised parameterisation of cation-binding by humic substances. *Environmental Chemistry*. 8(3):225–235. doi:10.1071/EN11016.
- Van Laer L, Smolders E, Degryse F, Janssen C, De Schamphelaere KAC. 2006. Speciation of nickel in surface waters measured with the Donnan membrane technique. *Analytica Chimica Acta*. 578(2):195–202. doi:10.1016/j.aca.2006.06.070.
- Weyers A, Vollmer G. 2000. Algal growth inhibition: effect of the choice of growth rate or biomass as endpoint on the classification and labelling of new substances notified in the EU. *Chemosphere*. 41:1007–1010.

Worms IAM, Wilkinson KJ. 2008. Determination of Ni²⁺ using an equilibrium ion exchange technique: Important chemical factors and applicability to environmental samples. *Analytica Chimica Acta*. 616(1):95–102. doi:10.1016/j.aca.2008.04.004.

Table 1: Conditions of the various chronic Ni toxicity tests that measured growth inhibition in juvenile *Lymnaea stagnalis*

	Crémazy et al. 2018	Niyogi et al. 2014	Nys et al. 2016	Schlekat et al. 2010
Test snails origin	University of British Columbia (in-house culture)	University of Miami (in-house culture)	Eggs shipped from Vrije Universiteit Amsterdam.	Eggs shipped from University of Miami
Light: dark cycle	16:8 light: dark	16:8 light: dark	12:12 light: dark	16:8 light: dark
Water T (°C)	25 ± 1	25 ± 1	20	24 ± 2
Culture food	Sweet potato, romaine lettuce	Sweet potato, romaine lettuce	Organic butterhead lettuce	Sweet potato, romaine lettuce
Test food	Romaine lettuce, <i>ad libitum</i>	Romaine lettuce, <i>ad libitum</i>	Organic butterhead lettuce, 25 mg L ⁻¹	Carrot, romaine lettuce, sweet potato, <i>ad libitum</i>
Test duration	14 d	21 d	28 d	30 d
Test initial snail age	2 – 3 wk old	7 – 8 d old	2 – 3 d old	< 24 h old

Test initial snail weight	5 – 11 mg	2 – 4 mg	Not measured	Not measured
Test snail density	2.5 snails/L	4 snails/L	2 snails/L	8.3 snails/L
Test water regime	Static renewal (every day)	Static renewal (every day)	Static renewal (3 times/week)	Static renewal (3 times/week)
Test water(s) type	Vancouver water + salts	Miami water	Artificial waters	Natural waters
Test Ni salt	NiCl ₂	NiCl ₂	NiCl ₂	NiCl ₂
Water equilibration	> 12 h	24 h	24 h	> 18 h
Snail measurement	Wet weight	Wet weight	Shell length	Wet weight
Reported growth based on	Growth rate	Biomass	Growth rate	Biomass

Table 2: Literature observed and predicted EC20s and EC50s (dissolved Ni concentration, in $\mu\text{g L}^{-1}$) for growth inhibition in juvenile *L. stagnalis*. The observed ECxs are given as means with their 95% confidence intervals, based on snail biomass or specific growth rate. ECx predictions were made with WHAM VII and a *Ceriodaphnia dubia* chronic Ni BLM (eq. 2) with adapted sensitivity constants for *L. stagnalis* from

Schlekat et al. (2010) ($Q_{50, \text{Lymnaea, biomass}}$) and Nys et al. (2016) ($Q_{20, \text{Lymnaea, SGR}}$). These data are also illustrated in Figure 1.

Study	Water	Effect concentration	Biomass		Specific Growth Rate	
			Observed EC _x	Predict ed EC _x	Observed EC _x	Predict ed EC _x
Crémazy et al. 2018	UBC water	EC20	98 [73 - 120]	-	150 [130 - 170]	41
		EC50	160 [140 - 190]	7.0	220 [200 - 240]	-
Niyogi et al. 2014	UM water	EC20	< 1.3	-	9.4	36
		EC50	1.5	7.8	> 17	-
Nys et al. 2016	Artificial water pH 8.1	EC20	34 [21 – 54]	-	32 [21 - 50]	35
		EC50	> 39	5.4	> 39	-
Nys et al. 2016	Artificial water pH 8.6	EC20	13 [8.7 – 20]	-	10 [6.5 -16]	21
		EC50	> 42	3.3	40 [26 - 62]	-
Schlekat et al. 2010	Calapooia River	EC20	1.6 [1.4 - 2.3]	-	7.3 [1.1 - 50]	48
		EC50	6.2 [2.7 - 7.2]	8.1	10 [4.7 - 23]	-
Schlekat et al. 2010	S. Platte River	EC20	27 [12 - 31]	-	48 [33 - 69]	120
		EC50	40 [34 - 44]	26	98 [80 - 120]	-
Schlekat et al. 2010	S. Platte River- pH amended	EC20	21 [6.9 - 62]	-	84 [55 - 130]	490
		EC50	78 [66 - 85]	91	160 [130 - 200]	-
Schlekat et al. 2010	S. Santiam River	EC20	6.3 [1.3 - 14]	-	21 [18 - 24]	49
		EC50	15 [8.2 -	9.8	34 [31 - 36]	

20]

Schlekat et al. 2010	Zollner Creek	EC20	1.9 [1.7 - 3.0]	-	52 [42 - 65]	160
		EC50	23 [12 - 36]	36	110 [92 - 120]	-

Table 3: The experiment ID, Ni concentration range and measurement(s) (3-h uptake fluxes (J_{int}), 7-d and 14-d Specific Growth Rate (7-d SGR, 14-d SGR)) for each snail source and water type combination tested in the four experiments carried out at UBC or UM in the present study.

Exp ID	[Ni] ($\mu\text{g L}^{-1}$)	UBC snails		UM snails	
		UBC water	UM water	UBC water	UM water
Exp#1 (at UBC)	40, 300	3-h J_{int}	3-h J_{int}	-	-
Exp#2 (at UBC)	0, 10, 250	7-d SGR	7-d SGR	-	-
Exp#3 (at UM)	0, 10, 250	-	7-d SGR, 14-d SGR	7-d SGR, 14-d SGR	7-d SGR, 14-d SGR
Exp#4 (at UBC)	13 conc. [0 - 290]	14-d SGR	14-d SGR	-	-

a Testing of water type effects
b Testing of snail population effects
c Testing of lab effects

Table 4: Composition of the two exposure waters tested in the present study. The composition of UM water (dechlorinated Miami city water) was measured in the present study and is given as mean $\pm$ SD, n = 3 (except for Cl and SO₄ concentrations which are nominal). The composition of UBC water (dechlorinated Vancouver city water + salts) was measured in Crémazy et al. (2018) and is given as mean $\pm$ SD, n = 252 (except for Cl and SO₄ concentrations which are nominal).

	UM water	UBC water
pH	7.79 $\pm$ 0.02	7.81 $\pm$ 0.20
DOC (mg of C L ⁻¹)	2.16 $\pm$ 0.21	0.76 $\pm$ 0.08
DIC (mg of C L ⁻¹)	10.00 $\pm$ 0.10	9.56 $\pm$ 0.06
[Ca] (mM)	0.454 $\pm$ 0.006	0.94 $\pm$ 0.01
[Mg] (mM)	0.154 $\pm$ 0.005	0.22 $\pm$ 0.02
[Na] (mM)	1.50 $\pm$ 0.05	1.7 $\pm$ 0.2
[K] (mM)	0.087 $\pm$ 0.007	0.054 $\pm$ 0.015
[Cl] (mM)	1.21	1
[SO ₄] (mM)	0.14	0.79

Figure 1: Predicted versus observed a) $EC_{50\text{biomass}}$ and b) $EC_{20\text{SGR}}$ (dissolved Ni concentration) for growth inhibition of *L. stagnalis* in literature studies. Predictions were made with WHAM VII and a *Ceriodaphnia dubia* chronic Ni BLM (eq. 2) with adapted sensitivity constants for *L. stagnalis* from Schlekot et al. (2010) (Q_{50} , *Lymnaea*, biomass) and Nys et al. 2016 (Q_{20} , *Lymnaea*, SGR). Plain lines represent perfect predictions. The dashed and dotted lines represent predictions within a factor 2 and 4 respectively, for easier assessment of deviation from the 1:1 line. These data are also reported in Table 2.

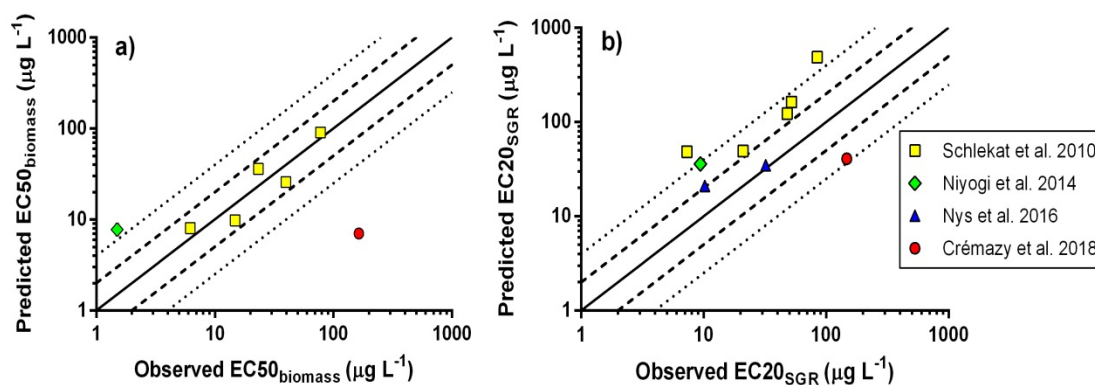


Figure 2: Free Ni^{2+} concentration (measured by an ion-exchange technique) as a function of total Ni concentration (nominal) in UBC and UM waters. Data are presented as mean $\pm$ SD ($n = 2$). The line corresponds to the 1:1 line.

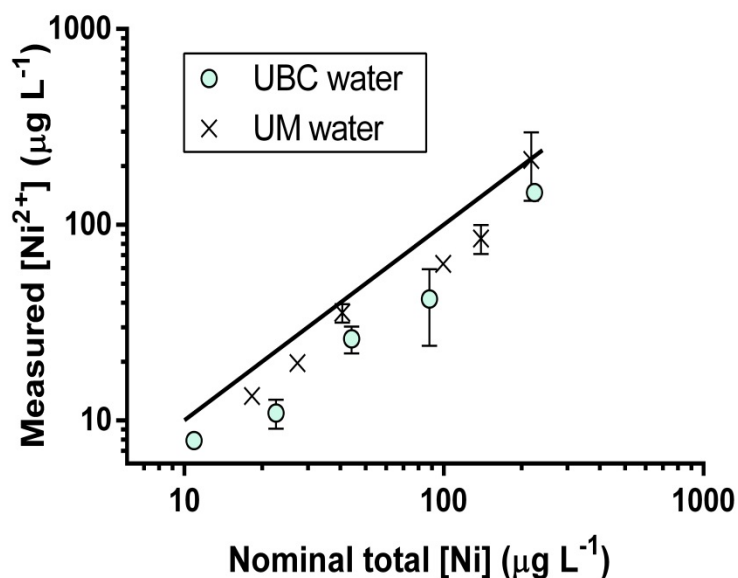


Figure 3:: Excitation-emission spectrum (wavelength in nm) of a) UM water, b) UBC water, and c) UBC water incubated with 40 $\mu\text{g L}^{-1}$ of total Ni.

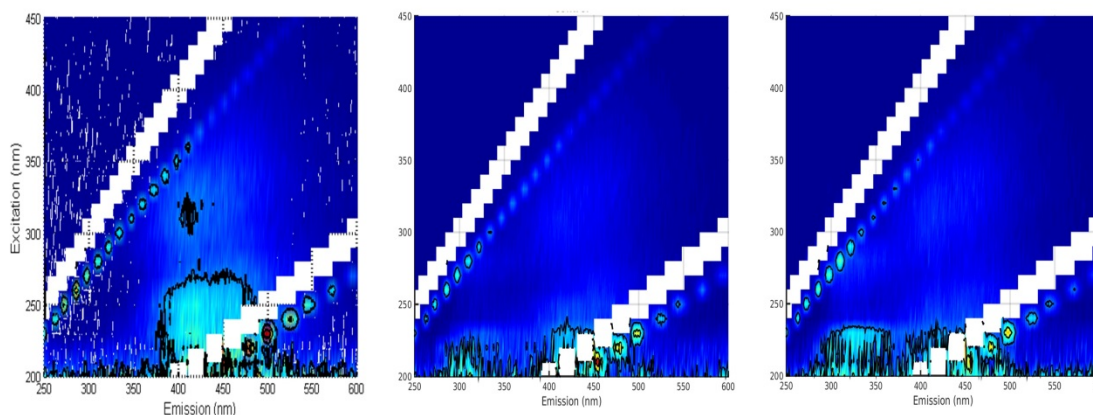


Figure 4: Effect of water type on Ni 3-h uptake rate in UBC snails, as a function of dissolved Ni concentration. Data are presented as means $\pm$ SE ($n = 6$). Results of the two-way ANOVA are given in each panel (ns: not significant, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$) and in more detail in Table SI.1 (Sup. Info). The interaction effect is abbreviated Int. Furthermore, statistically significant effects of [Ni] on Jint are shown by different letters (Tukey's test). No significant effect of water type on Jint was observed at each [Ni] with the Tukey's test.

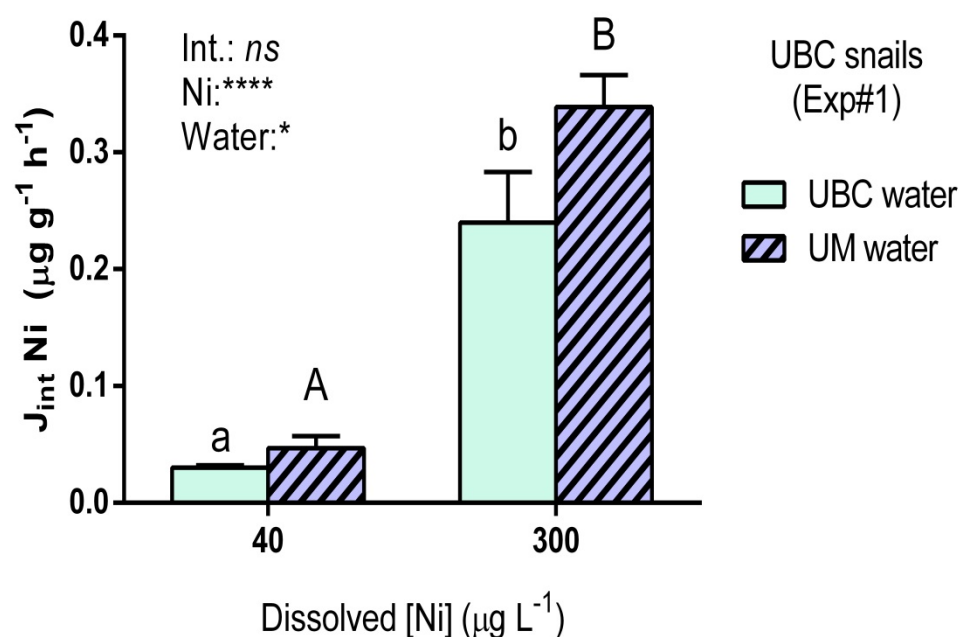


Figure 5: Effects of lab (a and a'), water type (with UBC snails: b and b'; with UM snails: c and c') and snail population (d and d', in UM water) on snail SGR (left panel) and wf (right panel), as a function of dissolved Ni concentration. Results are shown as means $\pm$ SE (n = 5). The two-way ANOVA results are given in each panel (ns: not significant, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, ****: $p < 0.0001$) and in more details in Table SI.1 (Sup. Info). The interaction effect is abbreviated Int. Furthermore, statistically significant effects of lab/water/snail on SGR/wf are shown by daggers and statistically significant effects of [Ni] on SGR/wf are shown by different letters (Tukey's test).

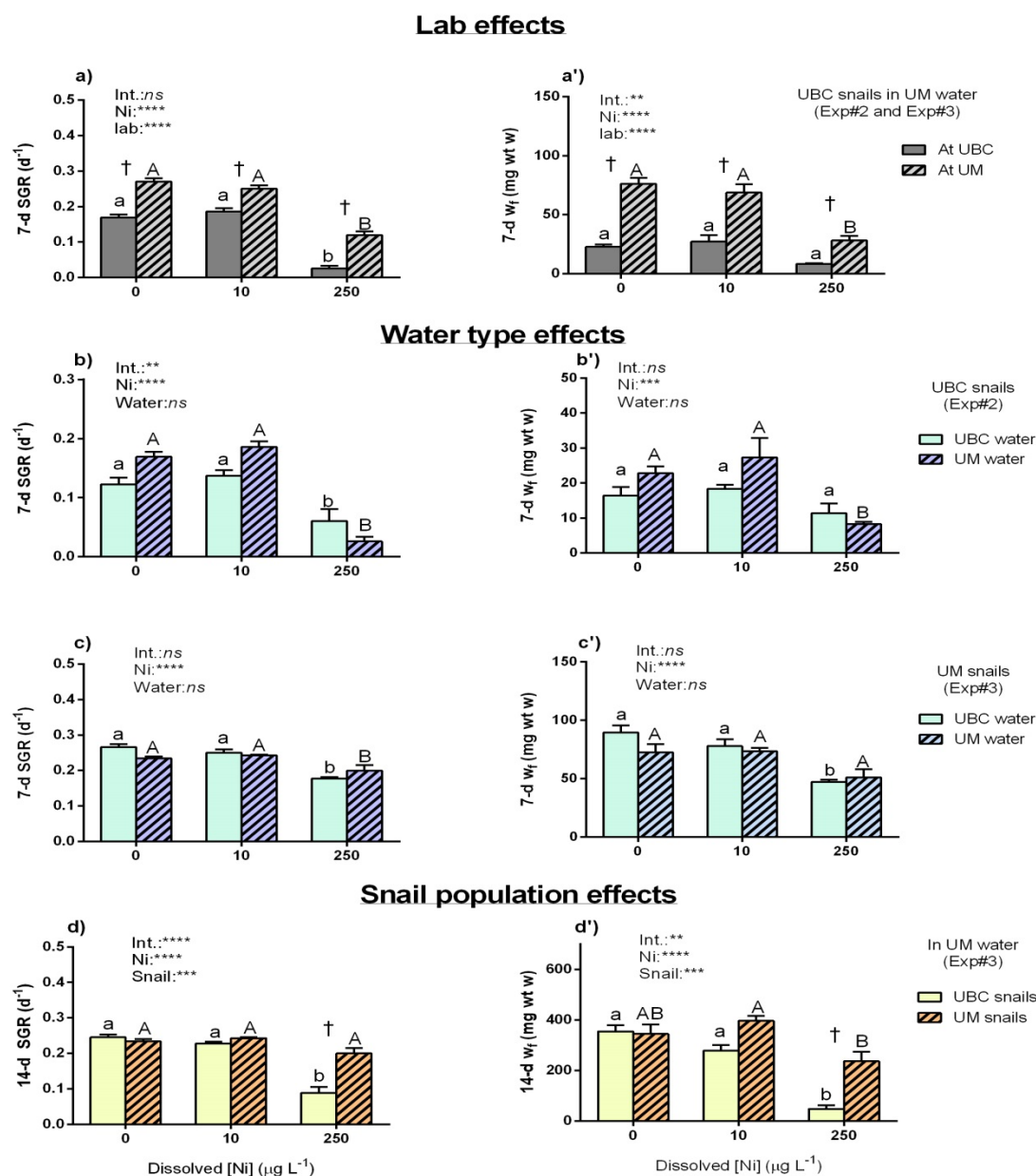


Figure 6: Effects of snail population (in UBC water) on snail a) SGR and a') wf, as a function of dissolved Ni concentration. Each data point corresponds to a treatment (n = 13 treatments). Regression and F-tests (**: $p < 0.01$, ***: $p < 0.001$) results are given in Table SI.2 (Sup. Info).

